Graphic Era HILL UNIVERSITY

Established by an Act of the State Legislature of Uttarakhand (Adhiniyam Sankhya 12 of 2011)
University under section 2(f) of UGC Act, 1956

Mid Term (Odd) Semester Examination October 2024

Roll no.....

Name of the Course and semester: B.TECH. CE and IIIrd
Name of the Paper: Geomatics Engineering
Paper Code: TCE-302
Time: 1.5 hour

Maximum Marks: 50

Note:
- *(i)* Answer all the questions by choosing any one of the sub questions
- *(ii)* Each question carries 10 marks.
- *(iii)* Please specify COs against each question.

Q1. (10 Marks)

a. Explain the methods of linear measurement and their relative accuracy. (CO1)

OR

b. A surveyor measures a distance of 200 meters using a 20-meter chain. If the chain is 5 cm too short, what is the correct length of the measured distance? (CO1)

Q2. (10 Marks)

a. What are the Methods of Traversing in Surveying? Explain the Procedure, and Advantages of Each Method. (CO2)

OR

b. In a compass traverse, a correction of 2° is applied to each bearing. Recomputed the bearings for initial values of 85°, 175°, 265°, and 355°. (CO2)

Q3. (10 Marks)

a. Explain the Principle, and Applications of Compass Traverse in Surveying. (CO2)

OR

b. In a triangulation survey, the base line is 1000 m, and the angles at the three stations are 45°, 60°, and 75°. Calculate the lengths of the other two sides. (CO2)

Q4. (10 Marks)

a. A steel tape of length 20 m is used to measure a distance at a temperature of 35°C. The tape was calibrated at 20°C. If the coefficient of thermal expansion of steel is 11×10^{-6} °C^{-}, calculate the correction for temperature. (CO1)

OR

b. Describe the process of chain surveying and its applications (CO1)

Q5. (10 Marks)

a. Discuss the principle and working of theodolite surveying. (CO1)

OR

b. A closed traverse consists of the following latitudes and departures: L1 = +50 m, L2 = -30 m, L3 = +40 m, D1 = +20 m, D2 = -10 m, D3 = +15 m. Compute the closing error. (CO2)